

Batch: C3 Roll No.: 16010123325
Experiment No. AWS -1
Grade: AA / AB / BB / BC / CC / CD /DD

Title: Launching and Managing EC2 Instances

Objective:

After completing this experiment, students will be able to:

- Understand basic cloud computing concepts
- Launch an EC2 instance
- Configure security groups
- Connect to an EC2 instance using SSH
- Install and configure a web server
- Monitor instance performance
- Manage instance lifecycle states

Expected Outcome of Experiment:

CO	Outcome
	To launch, configure, connect, monitor, and manage a virtual server using Amazon EC2 in AWS Cloud.

Books/ Journals/ Websites referred:

Abstract:-

Prerequisites

- AWS Account (Free Tier recommended)
- Basic knowledge of Linux commands
- Internet connection
- SSH client (Terminal / Git Bash / PuTTY)

Related Theory: -

5.1 Cloud Computing

Cloud computing provides on-demand delivery of computing resources such as servers, storage, and networking over the internet.

5.2 Amazon EC2

Amazon EC2 (Elastic Compute Cloud) is a web service that provides resizable virtual servers in the cloud.

5.3 Key Concepts

- AMI (Amazon Machine Image): Pre-configured operating system template
- Instance Type: Hardware configuration (CPU, RAM)
- Key Pair: Secure authentication mechanism
- Security Group: Virtual firewall controlling inbound/outbound traffic
- EBS Volume: Persistent storage attached to instance

Procedure:-

Step 1: Login to AWS Console

1. Open AWS Management Console
2. Search for EC2 service
3. Click “Launch Instance”

Step 2: Configure EC2 Instance

1. Name the instance: Lab2-EC2-Instance
2. Choose AMI: Amazon Linux 2023
3. Select Instance Type: t2.micro (Free Tier eligible)
4. Create a new Key Pair and download the .pem file
5. Configure Network Settings:

- Allow SSH (Port 22)
- Allow HTTP (Port 80)
- 6. Keep default storage (8 GB)
- 7. Click Launch Instance

Step 3: Connect to EC2 Instance

Method 1: Browser-Based Connection

- Select instance → Click Connect → EC2 Instance Connect

Method 2: SSH via Terminal

```
chmod 400 keypair.pem  
ssh -i keypair.pem ec2-user@<Public-IP>
```

Step 4: Install Apache Web Server

```
sudo yum update -y  
sudo yum install httpd -y  
sudo systemctl start httpd  
sudo systemctl enable httpd
```

Open browser and enter:

http://

Step 5: Monitoring the Instance

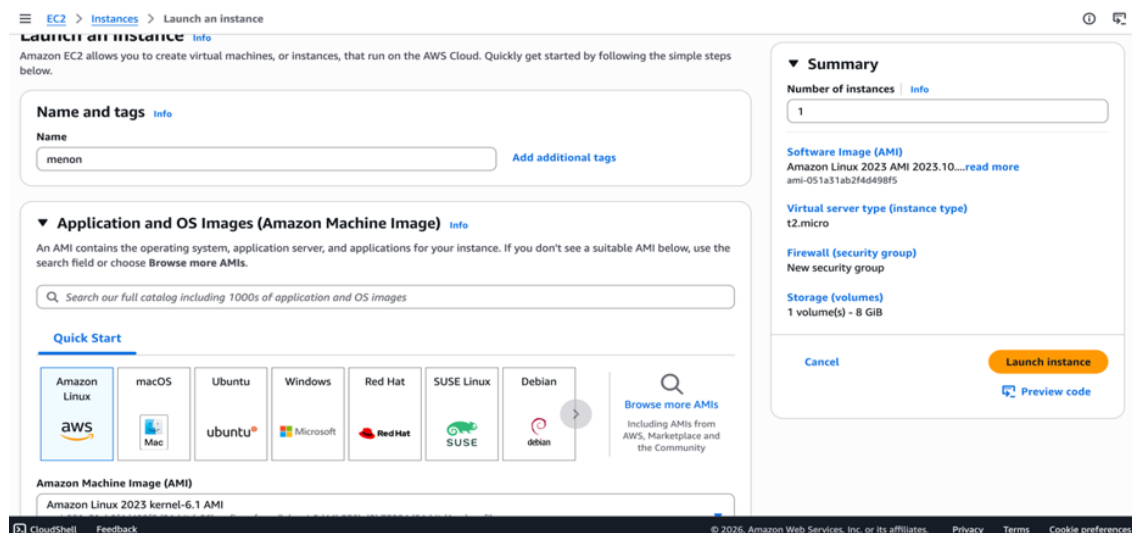
1. Go to EC2 Dashboard
2. Select Instance
3. Click Monitoring tab

Observe: - CPU Utilization - Network In/Out - Status Checks

Implementation Details:

Launch Instance

1. Name your instance : menon
2. Select AMI – Amazon Linux
3. Select Instance Type – t2.micro
4. Create a new key pair : RSA (smenon.pem)



The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA

RSA encrypted private and public key pair

☐ ED25519

ED25519 encrypted private and public key pair

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

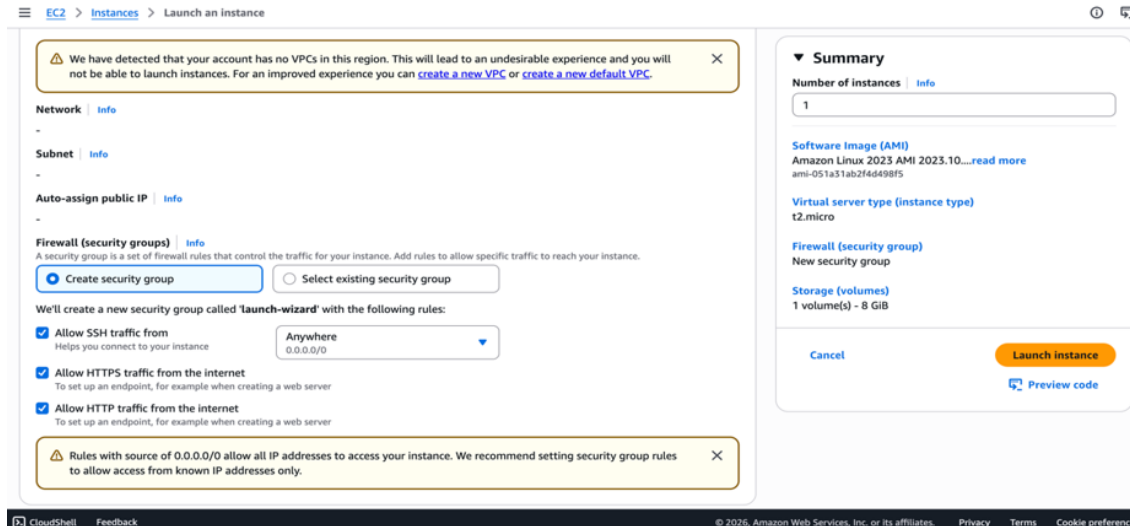


When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Edit your security group rules

1. Create security group

2. SSH traffic : Anywhere



The screenshot shows the 'Launch an instance' page in the AWS Management Console. A warning message at the top states: 'We have detected that your account has no VPCs in this region. This will lead to an undesirable experience and you will not be able to launch instances. For an improved experience you can [create a new VPC](#) or [create a new default VPC](#).' Below this, the 'Firewall (security groups)' section is expanded, showing the option to 'Create security group'. A new security group named 'launch-wizard' is being created with the following rules:

- ☒ Allow SSH traffic from: Anywhere (0.0.0.0/0)
- ☒ Allow HTTPS traffic from the internet
- ☒ Allow HTTP traffic from the internet

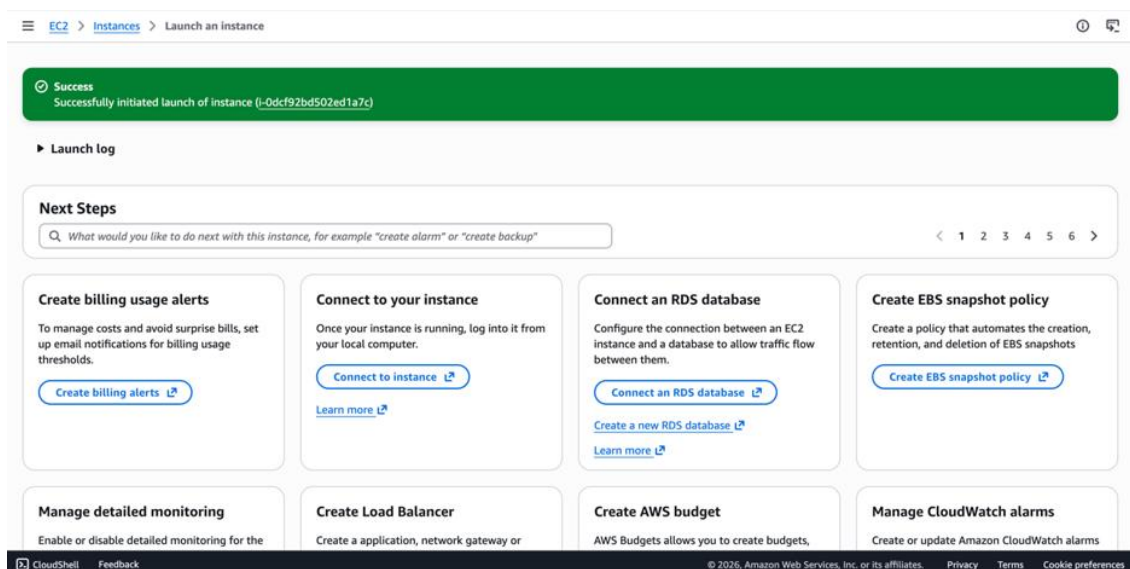
A warning message at the bottom of the rules section states: 'Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

The 'Summary' section on the right shows the following configuration:

- Number of instances: 1
- Software image (AMI): Amazon Linux 2023 AMI 2023.10...
- Virtual server type (instance type): t2.micro
- Firewall (security group): New security group
- Storage (volumes): 1 volume(s) - 8 GiB

Buttons for 'Cancel', 'Launch instance', and 'Preview code' are visible at the bottom right.

Launch Instance

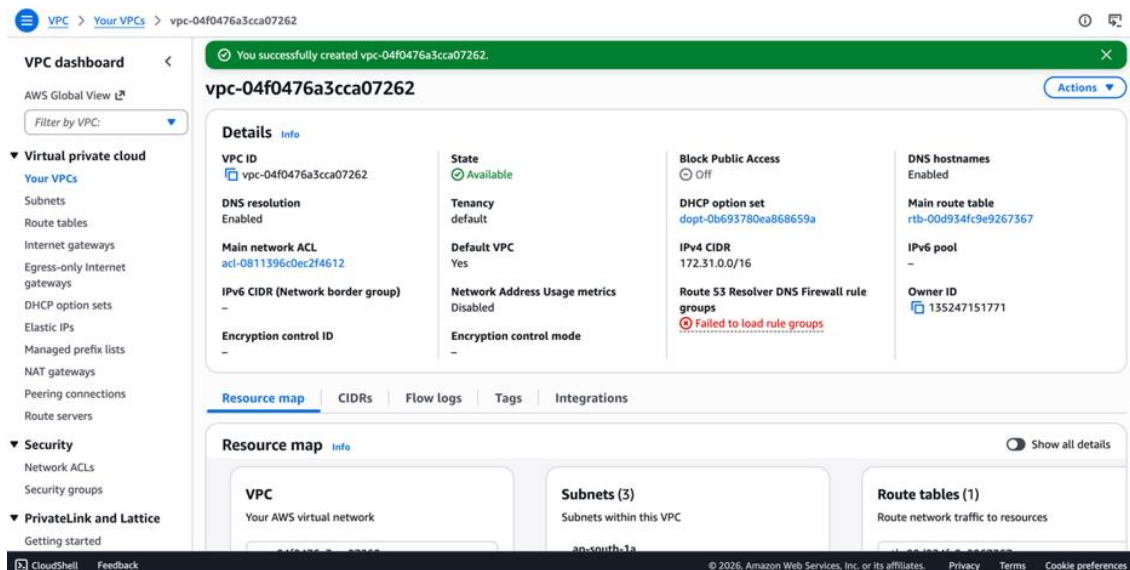


The screenshot shows the 'Launch an instance' page in the AWS Management Console. A green success message at the top states: 'Successfully initiated launch of instance (i-0dcf92bd502ed1a7c)'. Below this, the 'Next Steps' section is expanded, showing a search bar and a list of recommended actions:

- Create billing usage alerts**: To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds. [Create billing alerts](#)
- Connect to your instance**: Once your instance is running, log into it from your local computer. [Connect to instance](#)
- Connect an RDS database**: Configure the connection between an EC2 instance and a database to allow traffic flow between them. [Connect an RDS database](#)
- Create EBS snapshot policy**: Create a policy that automates the creation, retention, and deletion of EBS snapshots. [Create EBS snapshot policy](#)
- Manage detailed monitoring**: Enable or disable detailed monitoring for the instance. [Manage detailed monitoring](#)
- Create Load Balancer**: Create an application, network gateway or [Load Balancer](#)
- Create AWS budget**: AWS Budgets allows you to create budgets. [Create AWS budget](#)
- Manage CloudWatch alarms**: Create or update Amazon CloudWatch alarms. [Manage CloudWatch alarms](#)

Connect to EC2 Instance

Public IP- 13.203.24.100



Connect using Public IP : On device Terminal

1. Navigate to the folder where the .pem file is stored
2. Change Key Permission

```
Last login: Thu Mar 5 11:57:48 on ttys000
shreyas@Shreyas-MacBook-Air-2: ~ % cd Downloads
shreyas@Shreyas-MacBook-Air-2: Downloads % chmod 400 smenon.pem
shreyas@Shreyas-MacBook-Air-2: Downloads %
```

Connect to the EC2 Instance

```
Last login: Thu Mar 5 11:57:48 on ttys000
shreyas@Shreyas-MacBook-Air-2: ~ % cd Downloads
shreyas@Shreyas-MacBook-Air-2: Downloads % chmod 400 smenon.pem
shreyas@Shreyas-MacBook-Air-2: Downloads % ssh -i smenon.pem ec2-user@13.203.24.100
The authenticity of host '13.203.24.100 (13.203.24.100)' can't be established.
ED25519 key fingerprint is: SHA256:jfXki-c650hg9kzazK3Xoe+IvoI2rpd5Vg4kocDw0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.203.24.100' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-24-193 ~]$
```

Install Apache Web Server



```
Complete!
[ec2-user@ip-172-31-24-193 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:02:09 ago on Thu Mar  5 06:32:26 2026.
Dependencies resolved.

Package                               Architecture      Version           Repository        Size
Installing:
httpd                                 x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      47 k
Installing dependencies:
apr                                   x86_64            1.7.5-1.amzn2023.0.4  amazonlinux      129 k
apr-util                             x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      97 k
apr-util-ldap                        x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      13 k
generic-logos-httpd                 noarch            18.0.0-12.amzn2023.0.3  amazonlinux      19 k
httpd-core                           x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      1.4 M
httpdfilesystem                     noarch            2.4.66-1.amzn2023.0.1  amazonlinux      13 k
httpd-tools                         x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      81 k
libbrotli                           x86_64            1.0.9-4.amzn2023.0.2  amazonlinux      315 k
mailcap                             noarch            2.1.49-3.amzn2023.0.3  amazonlinux      33 k
Installing weak dependencies:
apr-util-openssl                    x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      15 k
mod_http2                           x86_64            2.0.27-1.amzn2023.0.3  amazonlinux      166 k
mod_lua                             x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      60 k

Transaction Summary
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                2.3 MB/s | 97 kB  00:00
(2/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm           307 kB/s | 13 kB  00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                    2.7 MB/s | 129 kB  00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm        677 kB/s | 15 kB  00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm   742 kB/s | 19 kB  00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm                 1.9 MB/s | 47 kB  00:00
(7/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64.rpm           4.1 MB/s | 81 kB  00:00
(8/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64.rpm            41 MB/s | 1.4 MB  00:00
(9/13): httpdfilesystem-2.4.66-1.amzn2023.0.1.noarch.rpm        414 kB/s | 13 kB  00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm             10 MB/s | 315 kB  00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm              1.3 MB/s | 33 kB  00:00
(12/13): mod_http2-2.0.27-1.amzn2023.0.3.x86_64.rpm            4.7 MB/s | 166 kB  00:00
(13/13): mod_lua-2.4.66-1.amzn2023.0.1.x86_64.rpm              2.2 MB/s | 60 kB  00:00
Total                                                            12 MB/s | 2.4 MB  00:00

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64                    1/1
Installing : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64          1/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64      2/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64      3/13
```

Verify Apache is Running

```
Complete!
[ec2-user@ip-172-31-24-193 ~]$
[ec2-user@ip-172-31-24-193 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-24-193 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service + /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-24-193 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-03-05 06:35:36 UTC; 58s ago
     Docs: man:httpd.service(8)
   Main PID: 25707 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1072)
    Memory: 12.1M
       CPU: 163ms
    CGroup: /system.slice/httpd.service
            └─25707 /usr/sbin/httpd -DFOREGROUND
            └─25708 /usr/sbin/httpd -DFOREGROUND
            └─25709 /usr/sbin/httpd -DFOREGROUND
            └─25710 /usr/sbin/httpd -DFOREGROUND
            └─25737 /usr/sbin/httpd -DFOREGROUND

Mar 05 06:35:36 ip-172-31-24-193.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 05 06:35:36 ip-172-31-24-193.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 05 06:35:36 ip-172-31-24-193.ap-south-1.compute.internal httpd[25707]: Server configured, listening on: port 80
[ec2-user@ip-172-31-24-193 ~]$
```

```
Running scriptlet: httpd filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 9/13
Installing : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 10/13
Installing : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.66-1.amzn2023.0.1.x86_64 12/13
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 3/13
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13
Verifying : httpd filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13
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Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13

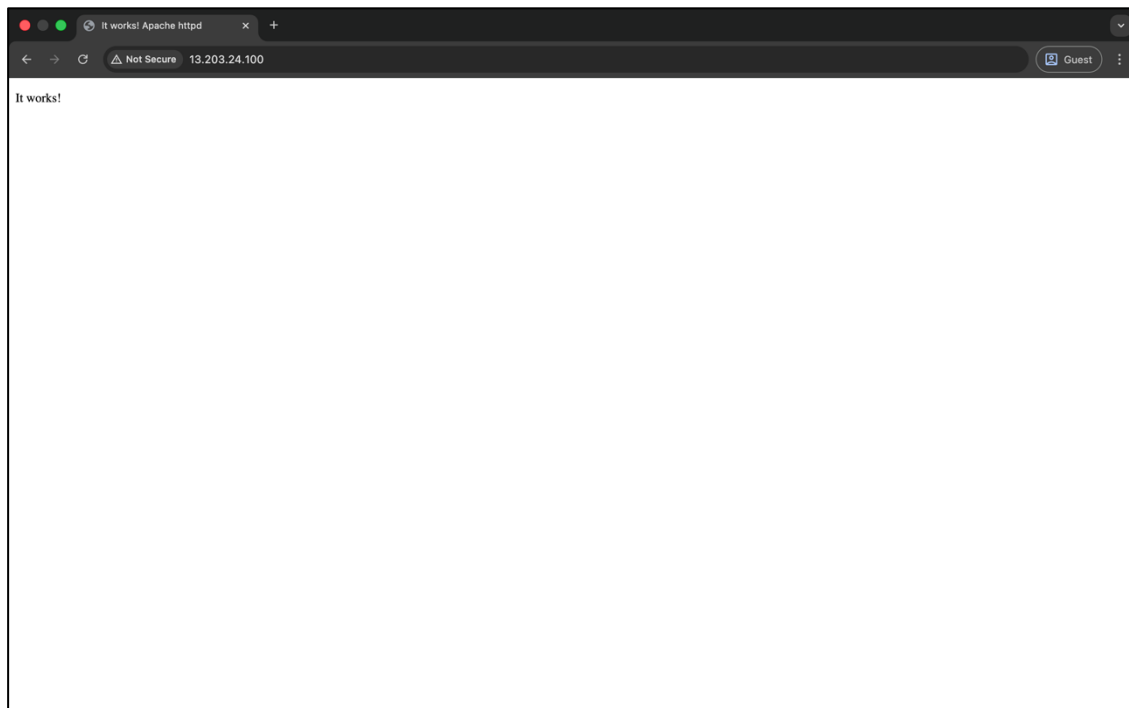
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.2.x86_64      apr-util-1.6.3-1.amzn2023.0.2.x86_64      apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.66-1.amzn2023.0.1.x86_64      httpd-core-2.4.66-1.amzn2023.0.1.x86_64      httpd filesystem-2.4.66-1.amzn2023.0.1.noarch
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64      libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-24-193 ~]$
[ec2-user@ip-172-31-24-193 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-24-193 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service + /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-24-193 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
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            └─25710 /usr/sbin/httpd -DFOREGROUND
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Mar 05 06:35:36 ip-172-31-24-193.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 05 06:35:36 ip-172-31-24-193.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 05 06:35:36 ip-172-31-24-193.ap-south-1.compute.internal httpd[25707]: Server configured, listening on: port 80
[ec2-user@ip-172-31-24-193 ~]$
[ec2-user@ip-172-31-24-193 ~]$
```

Output

Apache Test Page, which confirms that the web server is working



Post Lab:-

1. What is EC2?

Amazon EC2 (Elastic Compute Cloud) is a cloud computing service provided by Amazon Web Services that allows users to rent virtual servers (instances) to run applications on the cloud.
It provides scalable computing capacity without needing physical hardware.

2. What is the difference between Stop and Terminate?

Feature	Stop	Terminate
Meaning	Temporarily shuts down instance	Permanently deletes instance
Data	Data on EBS is retained	Data is lost (unless backed up)
Restart	Can restart anytime	Cannot restart
Billing	No compute charges (only storage)	No charges after deletion

3. What is AMI?

Amazon Machine Image (AMI) is a pre-configured template used to launch EC2 instances.

It includes:

- Operating system
- Application software
- Configuration settings

4. What is a Security Group?

A **Security Group** is a virtual firewall in AWS that controls incoming and outgoing traffic for EC2 instances.

It defines:

- Allowed IP addresses
- Ports (e.g., 22 for SSH, 80 for HTTP)
- Protocols

5. What is the purpose of a Key Pair?

A **Key Pair** is used for securely logging into EC2 instances.

It consists of:

- **Public key** → Stored on AWS
- **Private key** → Stored with user

6. What is monitoring in cloud computing?

Monitoring in cloud computing refers to tracking the performance, health, and usage of cloud resources.

In AWS, this is mainly done using **Amazon CloudWatch**.

It helps to:

- Track CPU usage, memory, network activity
- Detect failures or unusual behavior
- Set alerts (alarms)

Conclusion:-

This experiment demonstrated how virtual machines can be created and managed in a cloud environment using EC2 service. It provided hands-on experience in launching, configuring, and monitoring cloud-based infrastructure.